

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department: Computer Science and Engineering

Program: B.Sc. in Computer Science and Engineering

Semester Final Examination: Spring 2020

Year: 4th

Semester: 1st

Course Number: CSE4107

Course Name: Artificial Intelligence

Time: 3 (Three) Hours

Full Marks: 60

Use SEPARATE answer scripts for Part A and Part B.

Instructions	i)	Answer scripts should be handwritten and should be written in A4 white paper. You must submit the hard copy of the answer scripts to the Department when the university reopens.
	ii)	You must write the following information at the top page of each part : Department: Course no.: Examination: Student ID: Program: Course Name: Semester (Session): Signature and Date:
	iii)	Write down Student ID, Course number and put your signature on top of every single page of the answer script.
	iv)	Write down page number at the bottom of every page of the answer script.
	v)	Upload the scan copy of your answer script in PDF format through provided google form at the respective course site (i.e., google classroom) using institutional email within the allocated time. Uploading clear and readable scan copy (uncorrupted) is your responsibility and you must cover all the pages of your answer script. However, for clear and readable scan copy of the answer script student should use only one side of a page for answering the questions.
	vi)	You must avoid plagiarism and, maintain academic integrity and ethics . You are not allowed to take any help from another individual. If you do so, it can result in stern disciplinary actions against you from the university authority.
	vii)	Marks allotted are indicated in the right margin .
	viii)	Assume any reasonable data if needed.
	ix)	Symbols and characters have their usual meaning.

Part A

Instructions	i)	The answer script (Single PDF file for Part A) must be uploaded at designated location in the provided Google Form link available in the Google classroom.
	ii)	Before uploading, rename the PDF file as follows. CourseNo_StudentID_PartNo Example: CSE4107_160204018_PartA.pdf
	iii)	There are 4 (Four) questions in this part, answer any 3 (Three) .

Question 1. [Marks: 13 1/3]

a)	Think of an agent in the form of an autonomous vehicle that is supposed to roam on the rocky surface of an alien planet and to collect data. Illustrate justified structural and functional components of it.	[6]
b)	Analyze the requirements of a knowledgebase for applying the chaining algorithms on it. Also, prove or disprove the proposition M using a typical forward chaining algorithm on the knowledgebase that follows. A, B, C, D, $A \wedge B \Rightarrow R$, $A \wedge G \Rightarrow S$, $A \wedge P \Rightarrow E$, $B \wedge D \Rightarrow Q$, $B \wedge E \Rightarrow G$, $C \wedge Q \Rightarrow K$, $D \wedge F \Rightarrow M$, $D \wedge K \Rightarrow P$, $D \wedge P \Rightarrow F$, $P \wedge R \Rightarrow H$.	[2+5 1/3]

Question 2. [Marks: 13 1/3]

a)	Given in the following tables a weighted undirected graph along with the heuristic function values of the nodes, considering A as the initial node and G as the goal node. <table><tr><td>Node</td><td>A</td><td>A</td><td>B</td><td>B</td><td>B</td><td>C</td><td>C</td><td>D</td><td>D</td><td>E</td><td>E</td><td>F</td></tr><tr><td>Neighbor</td><td>B</td><td>C</td><td>C</td><td>D</td><td>E</td><td>E</td><td>F</td><td>E</td><td>G</td><td>F</td><td>G</td><td>G</td></tr><tr><td>Weight</td><td>46</td><td>54</td><td>47</td><td>36</td><td>25</td><td>35</td><td>32</td><td>68</td><td>62</td><td>57</td><td>58</td><td>45</td></tr></table> <table><tr><td>Node</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td></tr><tr><td>h(Node)</td><td>75</td><td>51</td><td>53</td><td>56</td><td>54</td><td>38</td><td>0</td></tr></table> Find a solution to the problem instance demonstrating execution of the major steps of a typical Greedy Best-First search algorithm.	Node	A	A	B	B	B	C	C	D	D	E	E	F	Neighbor	B	C	C	D	E	E	F	E	G	F	G	G	Weight	46	54	47	36	25	35	32	68	62	57	58	45	Node	A	B	C	D	E	F	G	h(Node)	75	51	53	56	54	38	0	[6]
Node	A	A	B	B	B	C	C	D	D	E	E	F																																													
Neighbor	B	C	C	D	E	E	F	E	G	F	G	G																																													
Weight	46	54	47	36	25	35	32	68	62	57	58	45																																													
Node	A	B	C	D	E	F	G																																																		
h(Node)	75	51	53	56	54	38	0																																																		
b)	Illustrate the odds of the environment of an optimization problem containing various types of local maxima in it. Explain with it a situation when a randomly generated successor of a current state having a comparatively lesser value than the other successors will be more promising.	[4+3 1/3]																																																							

Question 3. [Marks: 13 1/3]

a)	Why is Game Playing called an adversarial search problem? Explain the most important characteristics of a game playing problem with more than two players.	[2+5 1/3]
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b)	Consider a planning problem instance with the Air Cargo Transportation problem given below. <table border="1" style="border-collapse: collapse; margin: 10px;"> <tr><td>A1</td></tr> <tr><td>C1 C2</td></tr> <tr><td>P1 P2</td></tr> </table> <table border="1" style="border-collapse: collapse; margin: 10px;"> <tr><td>A2</td></tr> <tr><td>C3 C4</td></tr> <tr><td>P3 P4</td></tr> </table> <table border="1" style="border-collapse: collapse; margin: 10px;"> <tr><td>A3</td></tr> <tr><td>C5 C6</td></tr> <tr><td>P5 P6</td></tr> </table> <table border="1" style="border-collapse: collapse; margin: 10px;"> <tr><td>A1</td></tr> <tr><td>C4 C6</td></tr> </table> <table border="1" style="border-collapse: collapse; margin: 10px;"> <tr><td>A3</td></tr> <tr><td>C3 C1</td></tr> </table> Initial State Goal State Explain the process of executing some predefined actions for finding a solution to the instance.	A1	C1 C2	P1 P2	A2	C3 C4	P3 P4	A3	C5 C6	P5 P6	A1	C4 C6	A3	C3 C1	[6]
A1															
C1 C2															
P1 P2															
A2															
C3 C4															
P3 P4															
A3															
C5 C6															
P5 P6															
A1															
C4 C6															
A3															
C3 C1															

Question 4. [Marks: 13 ¹/₃]																							
a)	Full joint-probability distribution of two random variables, having three different values each is shown in the table below. The variables are assumed to describe the knowledgebase of a given domain uniquely. <table border="1" style="border-collapse: collapse;"> <tr> <th colspan="2" rowspan="2"></th><th colspan="3">X_2</th></tr> <tr> <th>v_{21}</th><th>v_{22}</th><th>v_{23}</th></tr> <tr> <th rowspan="3">X_1</th><th>v_{11}</th><td>0.05</td><td>0.08</td><td>0.13</td></tr> <tr> <th>v_{12}</th><td>0.21</td><td>0.14</td><td>0.12</td></tr> <tr> <th>v_{13}</th><td>0.09</td><td>0.15</td><td>0.03</td></tr> </table> Compute the probabilities of the compound propositions, $v_{12} \vee v_{23}$, $v_{12} v_{21}$ and $v_{22} v_{12}$ using the distribution. You need to apply the Bayes' rule to calculate one of the two conditional probabilities.			X_2			v_{21}	v_{22}	v_{23}	X_1	v_{11}	0.05	0.08	0.13	v_{12}	0.21	0.14	0.12	v_{13}	0.09	0.15	0.03	[6]
				X_2																			
		v_{21}	v_{22}	v_{23}																			
X_1	v_{11}	0.05	0.08	0.13																			
	v_{12}	0.21	0.14	0.12																			
	v_{13}	0.09	0.15	0.03																			
b)	Write an illustrative essay on a possible way of application of the basic principle of decision theory.	[7 ¹ / ₃]																					

Part B

Instructions	i)	The answer script (Single PDF file for Part B) must be uploaded at designated location in the provided Google Form link available in the Google classroom.
	ii)	Before uploading, rename the PDF file as follows. CourseNo_StudentID_PartNo Example: CSE4107_160204018_PartB.pdf
	iii)	There are <u>3 (Three)</u> questions in this part, answer <u>any 2 (Two)</u> .

Table 1: Tennis Playing Dataset

Outlook	Temp	Windy	Play
sunny	hot	TRUE	no
sunny	hot	FALSE	yes
overcast	hot	FALSE	no
rainy	mild	FALSE	yes
sunny	cool	TRUE	yes
overcast	cool	TRUE	yes
overcast	cool	FALSE	yes
rainy	mild	FALSE	yes
rainy	cool	TRUE	no
rainy	mild	FALSE	no
overcast	mild	FALSE	yes
sunny	mild	FALSE	yes
sunny	hot	TRUE	yes

Table 2: Laptop Buying Dataset

age	student	income	rating	buy_laptop
26...35	yes	low	excellent	yes
<=25	no	medium	fair	no
<=25	yes	low	fair	yes
>35	yes	medium	fair	yes
<=25	yes	medium	excellent	yes
26...35	no	medium	excellent	yes
26...35	yes	high	fair	yes
<=25	no	high	fair	no
<=25	no	high	excellent	no
26...35	no	high	fair	yes
>35	no	medium	fair	yes
>35	yes	low	fair	yes
>35	yes	low	excellent	no
>35	no	medium	excellent	no

Question 1. [Marks: 10]

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|----|---|-----|
| a) | Consider the <i>Tennis Playing Dataset</i> (Table 1). Find the root node and draw the decision tree by following the Iterative Dichotomiser 3 algorithm. | [6] |
| b) | A binary dataset contains five different continuous features based on real-life scenario. Assume that
(i) all the data are evenly distributed between two distinct classes and,
(ii) all the data are of one class.
Discuss the entropy and information gain based on these two assumptions. | [4] |

Question 2. [Marks: 10]

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|----|---|-----|
| a) | For a multivariate dataset derive and minimize the sum of squared error equation in linear regression method. If the last two digits of your Student ID is characterized as 'A', then find and sketch the best-fitting line equation for the data points (A, A+1), (A+1, A+2), (A+3, A+6), (A+4, A+4), (A+6, A+10). | [6] |
| b) | Discuss and find the difference among R-squared, MAE, MSE, and RMSE errors with an example. Which type of Naïve Bayes Model will efficiently work on categorical valued dataset and why? | [4] |

Question 3. [Marks: 10]

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|----|--|-----|
| a) | Consider the <i>Laptop Buying Dataset</i> (Table 2). Find the class (<i>yes</i> or <i>no</i>) for an X-year-old student whose earning falls in <i>medium</i> category and who has an <i>excellent</i> rating. Here, X = last two digits of your student ID, but if X < 20, then X = last two digits of your ID + 20. | [6] |
|----|--|-----|

b)

Find the number of clusters and their features by applying the K-means clustering algorithm. The datapoints are characterized in Table 3 below. Also, to determine K in K-means clustering algorithm for working with unbiased data, which process will you adopt (Elbow or Silhouette) and why?

Table 3: Sample Dataset

		Datapoints	
		X_1	X_2
Features	A	5	3
	B	6	8
	C	3	8
	D	4	2
	E	7	0